Yuxuan Gu

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EDUCATION

MASTER OF ELECTRICAL AND INFORMAION ENGINEERING **2021-2025**

Imperial College London, UK

- Dean's list for Year 2 (2022-2023)
- Dean's list for Year 1 (2021-2022) (top 5%)
- Modules: Circuits analysis, Analogue electronics, Digital electronics, Computer architecture, Control Systems, Signals and Communication, Programming, Mathematics
- Demonstrated good time management skills, investigating literature involving control, robotics, electronics, and telecommunication in extra-curricular time

A LEVELS & IGCSE **2018-2021**

Jinling High School, Nanjing, China

- Accumulative marks Top 1 in China – November 2020 Cambridge Outstanding Learner Award (Best Across Three) and gave a speech as a Student Rep in the awards ceremony
- Further Mathematics (A*), Physics(A*), Chemistry(A*), Mathematics(A*)

WORK EXPERIENCE

Undergraduate Research Opportunities Programme (UROP) **July-Sept 2023**

Imperial College London, UK

- Developed a colour-tracking 4-DOF robotic arm using remote control within ROS2 framework
- Integrated a USB camera as a sensor within the feedback loop for effective bus servo motor management, while also employing geometric inverse kinematics analysis to calculate joint variables from end-effector position and orientation
- Employed a Raspberry Pi to transmit commands to the motors through UART, while concurrently publishing image frames to ROS2 topics
- Implemented controller separation onto a remote machine, enabling communication with the local node through the ROS2 network, and subsequently conducted an analysis of stability margins
- Enhanced problem-solving skills and bolstered confidence through independent exploration

Undergraduate Teaching Assistant (UTA) **Sept 2023**

Imperial College London, UK

- Provided comprehensive assistance to first-year student during the lectures and problem classes

Software Engineer Internship **July-Sept 2022**

Evotrack, London, UK

- Analysed usage data of E-vehicles charging stations in Paris and ran several k-means clustering algorithms to divide stations into clusters
- Established a routine to obtain the number of given list of facilities around given addresses using Google Maps API
- Trained ensemble models (e.g., Gradient Boosting) to predict the future utilisation of a station based on the past usage data and number of surrounding facilities and achieved high accuracy
- Developed perseverance and motivation by self-directed exploration of challenging tasks

Work Shadowing **July-Sept 2023**

Uptime, London, UK

- Examined water pumping systems, analysed pressure signals, PID feedback control, and CHP units while improving communication skills through industry advancement discussions with the engineer

PROJECTS

Self-balancing autonomous maze-solving rover **June 2023**

Imperial College London, UK

- Developed a self-balancing rover that autonomously solves mazes, maps them in real-time, and identifies the shortest path
- Implemented a comprehensive system equipped with various components, such as NEMA 17 stepper motors, A4988 motor driver modules, an MPU-6050 gyroscope, an ESP32 microcontroller, an FPGA board, and a camera
- Leveraged FPGA vision technology to enhance image processing and analysis capabilities, enabling precise navigation and obstacle detection.
- Implemented sophisticated algorithms such as PID control, triangulation and A* algorithm, clustering methods, and graph theory to optimize rover performance and achieve efficient maze-solving strategies.
- Cultivated leadership within a six-people team, taking charge of FPGA vision, motor driving, PID control, and facilitating seamless communication between esp32/PC and FPGA/esp32

Remotely controlled rover **June 2022**

Imperial College London, UK

- Designed a rover that can identify rocks emitting unusual electromagnetic and acoustic signals
- Utilised Fourier Transform, filtering methods and modulation & demodulation techniques
- Sharpened practical and programming skills by debugging faults in circuits and Arduino codes
- Strengthened negotiation and persuasion skills through collaboration with five teammates and organizing regular meetings

AI-based plants water shortage warning device **July-Sept 2020**

Verimake, Nanjing, China

- Developed research methodologies by exploiting online resources
- Developed a model to match the image of leaves with soil humidity
- Came up with the idea of using AI to remind ordinary people to water the plant
- Collected image of leaves and soil humidity data using Raspberry Pi and Arduino

TECHNICAL SKILLS

- Programming language: C++, Python
- Simulation tools: MATLAB/Simulink, LTSpice
- Embedded system and robotics: Arduino, Raspberry Pi, Robot Operating System (ROS)
- Language: English (IELTS: overall 7.5 with each band no less than 7.0), Chinese (native)

EXTRA-CURRICULAR INTERESTS

- Active member of Imperial Badminton Club, attending social sessions and patiently teaching beginners.
- Active member of Imperial Chamber Music Society, featuring violin solos in concerts (E.g. 2022 Christmas Concert)
- Donated 500RMB to the Project Hope to support the anti-virus battle against COVID in 2020